Disease Propagation Simulation using SIR Model

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I. *Introduction:*

Herd immunity describes a situation where disease is unable to spread because a large part of the population is immune to that disease. People can be immune when they get vaccinated or recover after infection. When a lot of people are immune it is harder for the disease to spread because there are not many people who can catch it. So, the disease does not spread much, and it goes away faster.

The goal of this project was to investigate about herd immunity by using a disease simulation. The simulation follows a simplified SIR model where each person can be susceptible, infectious, recovered, or vaccinated. The main objective was to study how different vaccination rates affect the final number of recovered people and to compare the simulated herd immunity threshold with the theoretical threshold.

II. *Methodology:*

Each person in the population has one of four possible states: susceptible, infectious, recovered, or vaccinated. Susceptible persons can become infected, infectious persons can spread the disease, recovered persons are immune, and vaccinated persons are protected from infection.

The final experiment used the following values:

Disease duration: 7 days
Transmissibility: 0.2
Contacts per infectious person per day: 6
Population size: 30000
Simulation runs per vaccination rate: 5

The experiment was performed using vaccination rates from **0% to 100%** in steps of **10%**. Each simulation started with one infectious person, called patient zero. One timestep represents one day. During each timestep, every infectious person contact six randomly selected persons. If a contacted person is susceptible, infection occurs according to the transmissibility value. The simulation stops when no infectious persons remain.
For every vaccination rate, multiple simulation runs were performed. The mean and standard deviation were calculated for the final number of recovered persons and the total number of timesteps until disease extinction.

The expected herd immunity threshold was calculated using:

$$\mathbf{R_0 = D \times \beta \times C}$$

where:
D = disease duration
β = transmissibility
C = contacts per infectious person per day

Using the assigned values:
$\mathbf{R_0 = 7 \times 0.2 \times 6 = 8.4}$

The expected herd immunity threshold is:

$H = 1 - 1/R_0$

$H = 1 - 1/8.4 = 0.881 \approx 88.1\%$

Therefore, the theoretical herd immunity threshold is approximately **88.1% vaccination**.

III. *Results:*

The simulation results show a clear reduction in the number of recovered persons as the vaccination rate increases. At low vaccination rates, almost the entire non-vaccinated population becomes infected and later recovers. At higher vaccination rates, the disease has fewer susceptible persons available, and the outbreak becomes much smaller.

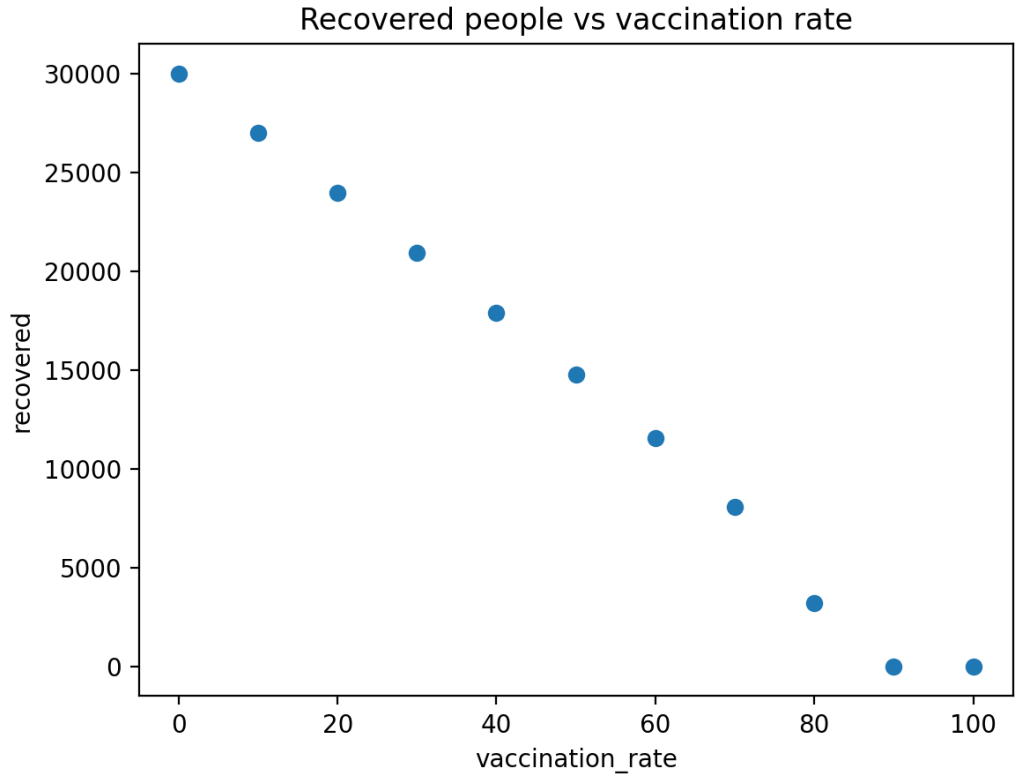


Figure 1: Simulation for $D = 7$, $C = 6$, $\beta = 0.1$.

Fig 1 shows that the final number of recovered persons decreases as the vaccination rate increases. The most important change occurs between **80% and 90% vaccination**, where the disease spread reduces sharply.

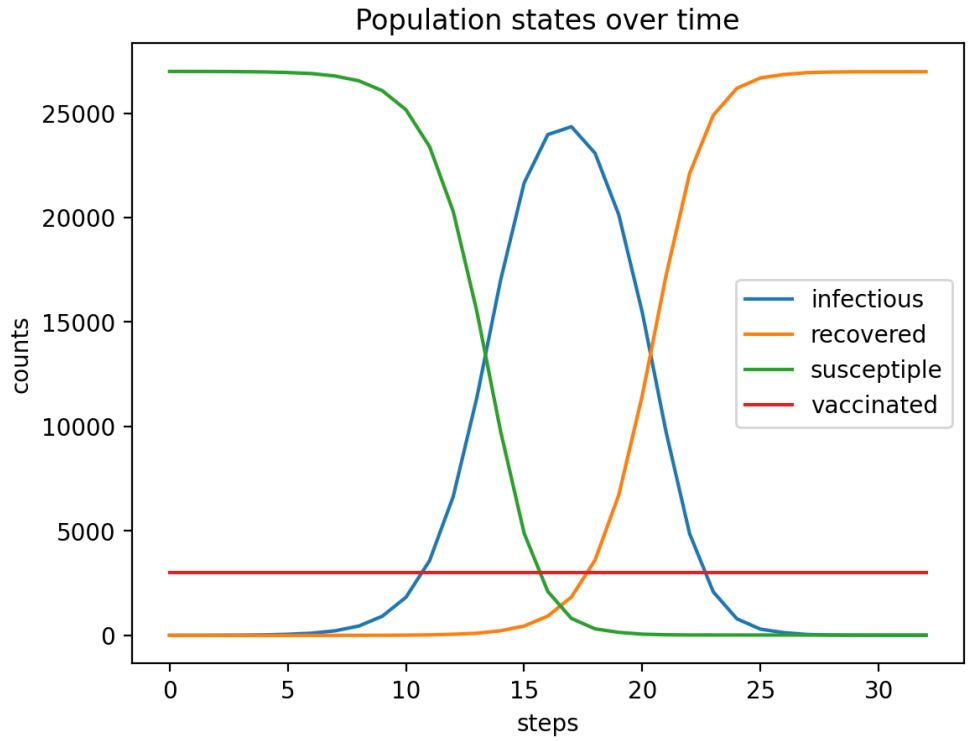


Figure 2: Timestep plot for $D = 7$, $C = 6$, $\beta = 0.1$ with vaccination rate of $V = 0.2$.

Fig 2 shows the timestep development of the disease states for a single simulation. The number of infectious persons first increases as the disease spreads, then decreases as infectious persons recover. The recovered count increases over time, while the susceptible count decreases during the outbreak.

IV. Discussion:

The results demonstrate the herd immunity effect clearly. The disease spreads through most of the susceptible population when the vaccination rate is low. As the vaccination rate increases fewer susceptible persons are available, so the final outbreak size becomes smaller.

The theoretical threshold was approximately 88.1%, and the simulation showed strong suppression of the disease at around 90% vaccination. These results are similar. The small difference can be explained by the stochastic nature of the simulation. Contacts are selected randomly, infections occur with probability, and only five simulation runs were performed for each vaccination rate.

Another reason for difference is that the vaccination rates were tested only in steps of 10%. Since the experiment tested 80% and 90% but not values such as 86%, 88%, or 89%, the exact simulated threshold cannot be identified more precisely. A smaller step size and more simulation runs would give a more accurate estimate.

The high standard deviation at 80% vaccination is also meaningful. It shows that the system is close to the threshold region. Near the threshold, small random differences between runs can strongly affect whether the disease spreads significantly or dies out quickly.

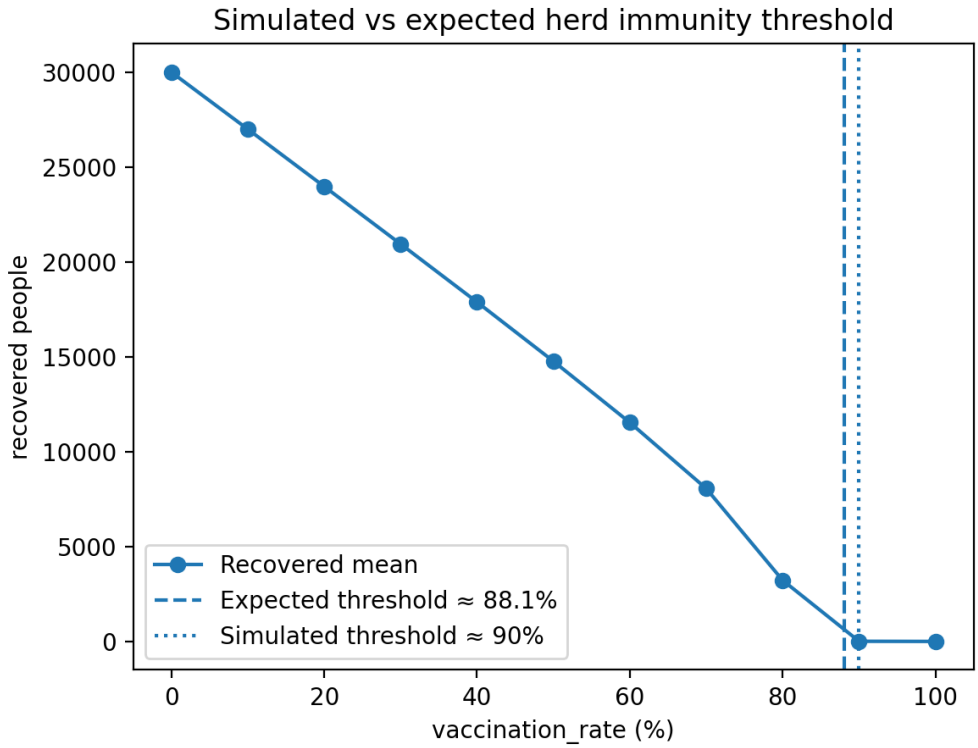


Figure 3: comparison of simulated threshold with the expected threshold

V. Conclusion:

For this project I investigated herd immunity by simulating the spread of disease between people. The last experiment was conducted with a population size of 30000 people, the disease lasted 7 days, had a transmissibility of 0.2 and vaccination rates between 0% and 100%.

This experiment demonstrates that the results of an increase in vaccination strongly reduces disease spread. The biggest change was when vaccination went from 80 to 90 percent. The predicted herd immunity threshold was calculated to be about 88.1% and the simulation showed that the disease was almost completely suppressed at 90% vaccination.

Overall, the simulation results are generally consistent with the theoretical expectation. The experiment shows the importance of herd immunity in controlling disease. If the rate of vaccination is high enough,a large outbreak can be prevented.